



Template-assisted electrodeposition of freestanding antimony, tin, and antimony-tin nanowire arrays from an ionic liquid

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Abstract

In the present study, we show the electrochemical synthesis of Sb, Sn, and Sb-Sn nanowire arrays from the ionic liquid 1-butyl-1-methylpyrrolidinium bis(trifluoromethylsulfonate) ([Py_{1,4}][TfO]) via templated-assisted approaches. Commercially available track etched polycarbonate template with a nominal pore diameter of 400 nm was utilized as a template. The nanowires were electrochemically deposited inside the pores of the template; then, a supporting copper layer was electrodeposited on the back side of the template. Subsequently, the template was dissolved with dichloromethane, and the structural morphology of the nanowire structures was explored by high-resolution scanning electron microscopy (SEM) and energy dispersive x-ray (EDX). Freestanding, mechanically stable nanowire arrays of Sb, Sn, and Sb-Sn with an average pore diameter of 400 nm were obtained. The charge/discharge characteristics of the electrodeposited nanowire films were investigated to explore the Li storage capacity of the fabricated electrodes. The results revealed that the electrodeposited nanowire films are promising anode candidates for the future generation of Li-ion batteries.

Keywords Electrodeposition · Ionic liquids · Nanowires · Li-ion batteries

Introduction

Alternative novel nanostructured anode materials have attracted considerable scientific interest to replace conventional graphite anodes in lithium-ion batteries to improve their performance as anodes for future generations of Li-ion batteries [1–5]. Metals and semiconductors, such as Sn, Sb, or Si-based materials, which can form alloys with lithium, are promising anode candidates [6–8]. This type of electrodes can have a considerably higher capacity over 900 mAh/g for tin [9] and 660 mAh/g for antimony [10] compared to graphite anode which has a theoretical capacity of only 372 mAh/g [11]. A noteworthy shortcoming

that hinders their utilization is the large volume changes which occur during cycling [8]. Such volume change results in cracking, which leads to contact loss and disintegration of the active material. Various strategies have been suggested to resolve this problem and improve the cyclability of the alloyed anodes. One of these strategies is to design the anode structure in the form of nanowire arrays to accommodate the large volume changes on cycling [12–15].

Various approaches for the synthesis of the nanowire structures have been reported, such as chemical vapor deposition, thermal spray, and electrochemical deposition [16–19]. Nevertheless, these methods are either multi-step or non-cost effective, making them less attractive for up-scaling. Electrodeposition is a beneficial approach as it is scalable and the experimental setup is less demanding [20]. Template-assisted electrodeposition is a smart approach for the synthesis of nanostructures with designed shapes such as freestanding nanowires [21–23]. The most commonly used membranes to synthesize nanowires are anodic aluminum oxide (AAO) and nuclear track-etched polycarbonate (PC). Tin, antimony, and tin–antimony (SnSb) alloys are supposed to be promising anodes for future generations of Li-ion batteries [24–27].

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We previously reported the electrodeposition of tin films from TfO-based ionic liquids under open air conditions [28]. Antimony nanowires were prepared by various routes such as chemical vapor deposition [29], focused ion beam induced synthesis [30], displacement reaction [31], hydrothermal reduction method [32], and template-free and template-assisted electrodeposition [8, 33, 34]. Quite recently, Endres et al. [35] have reported the electrochemical synthesis of antimony nanowires from the ionic liquid 1-butyl-1-methylpyrrolidinium bis(trifluoromethylsulfonyl) imide ($\text{Py}_{1,4}\text{TFSI}$) containing 0.5 mol/L SbCl_3 at room temperature without template. The electrodeposition was carried out in inert gas atmosphere [35].

It is worth noting that most of the electrodeposition processes from ionic liquids were conducted under inert gas conditions. In the present study, we report on the template-assisted electrodeposition of Sn, Sb, and SbSn nanowire architectures on an electrodeposited Cu as supporting layers in $[\text{Py}_{1,4}]\text{TfO}$ under open air conditions. We show herein that strongly adherent, stable freestanding nanowire arrays are obtained. Although Sb, Sn, and SbSn nanowires can be electrochemically made using aqueous electrolytes, employing ionic liquids in the template-assisted electrodeposition for nanowire structures can offer more advantages. It is well known that ionic liquids have a lower surface tension than water; therefore, a complete filling of the pores of the template can easily be reached using ionic liquid electrolytes which, in turn, leads to obtain homogenous nanowire structures after the electrodeposition process.

Experimental

The electrodeposition experiments were performed using the ionic liquid $[\text{Py}_{1,4}]\text{TfO}$ which was purchased from Io.Li. Tec., Germany. All experiments were conducted under open air conditions. Anhydrous SnCl_2 (PROLABO, 99%) and SbCl_3 (Acros) was used as a source of tin and antimony, respectively. The concentration of SnCl_2 and SbCl_3 in $[\text{Py}_{1,4}]\text{TfO}$ was 0.1 M. Track-etched polycarbonate (PC) membranes with a nominal thickness and an average pore diameter of 21 μm and 400 nm, respectively, were used as templates. The PC membranes with pore sizes of 400 nm were employed in this work in order to get freestanding, mechanically stable nanowire arrays. A thin gold film with thickness of about 50 nm was sputtered onto the PC template serving as the working electrode at room temperature. Both counter and reference electrodes were platinum wires (Alfa, 99.99%). After the electrodeposition of nanowires inside the membrane, a Cu supporting layer was electrodeposited on the Au-sputtered side of the membrane. For Cu electrodeposition experiments, copper wires (99.9% (metals basis), Thermo Scientific Chemicals) were utilized as counter and

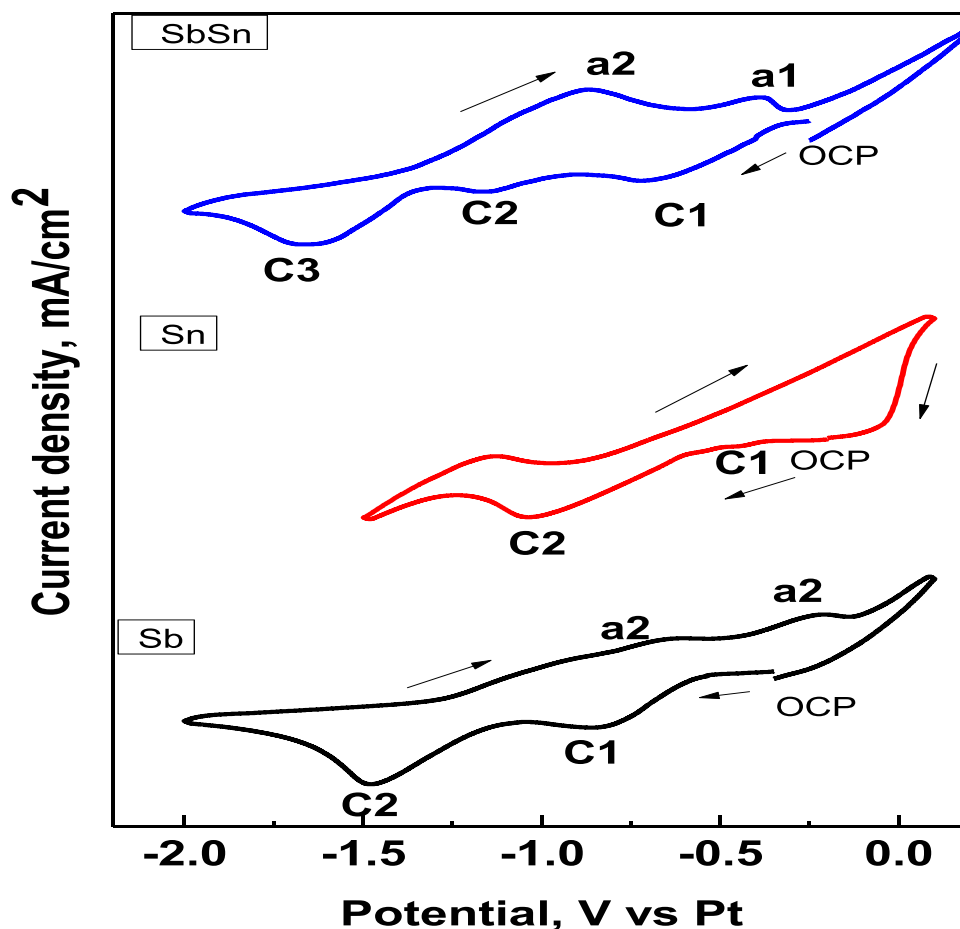
quasi-reference electrodes, respectively. An acidic aqueous solution of 0.6 M CuCl_2 (Alfa Aeser), pH 4, was employed as an electrolyte for the electrodeposition of a Cu supporting layer. The electrodeposition was conducted galvanostatically at a current density of -6 mA/cm^2 for 30 min. The supporting Cu layer was electrodeposited on the Au-sputtered side of the membrane, while the non-sputtered side was entirely separated from the electrolyte using Teflon tape.

After deposition experiments, the PC templates were subtracted by dissolving in dichloromethane (CH_2Cl_2 , AR, Merck). A Voltalab 40 Potentiostat/Galvanostat controlled by voltamaster 4 software was used in all electrochemical experiments. The morphology of the deposits was investigated using a field emission scanning electron microscope (Quanta FEG 250) with energy dispersive X-ray analyzer. A PANalytical diffractometer with CuK_α radiation was utilized to explore the phase composition of the deposits. The charge/discharge performance of the electrodeposited nanowire electrodes was investigated using a TEFLON cell where Li foil acts as both counter and reference electrodes. Lithium trifluoromethylsulphonyl imide dissolved in the ionic liquid $\text{LiTfSA}/[\text{Py}_{1,4}]\text{TfSA}$ was used as an electrolyte. The potential window was between 0.01 and 2.0 V vs. Li.

Results and discussion

The cyclic voltammetry behavior of 0.1 M SbCl_3 , 0.1 M SnCl_2 , and of 0.1 M $\text{SbCl}_3 + 0.1 \text{ M SnCl}_2$ in $[\text{Py}_{1,4}]\text{TfO}$ recorded on copper substrates, without using PC templates, was investigated, Fig. 1. Initially, the potential was scanned from (OCP) to -2 V (vs. Pt) in the negative direction, and then up to $+0.1 \text{ V}$ in the anodic scan, to avoid oxidation of the copper substrate, with a rate of 10 mV s^{-1} . The cyclic voltammogram of 0.1 M SbCl_3 in $[\text{Py}_{1,4}]\text{TfO}$ shows two reduction processes (c_1 and c_2) on the cathodic branch of the CV. The first reduction process, recorded at -0.6 V (c_1), can be correlated to underpotential deposition of Sb on copper or due to the alloying of copper with antimony. The second process (c_2), recorded at around -1.6 V , is attributed to the bulk deposition of Sb as a grey deposit was formed on the surface of copper substrate. Similar behavior was shown by Liu et al. [35] for the ionic liquid electrolyte 0.5 M $\text{SbCl}_3/[\text{Py}_{1,4}]\text{TFSI}$ on gold. It was shown that the cathodic wave recorded prior to the bulk deposition of Sb was attributable to the formation of Sb-Au alloy [35]. The corresponding anodic peaks (a_1 and a_2) are recorded on the anodic side of the CV. The CV of 0.1 M SnCl_2 in $[\text{Py}_{1,4}]\text{TfO}$ exhibits two reduction processes in the cathodic regime. The cathodic waves c_1 and c_2 might be attributable to the under potential deposition or due to formation of Cu-Sn alloy. The second reduction peak is correlated to the bulk deposition of Sn [36]. As a result of the underpotential deposition processes,

Fig. 1 Cyclic voltammograms of 0.1 M SbCl_3 , 0.1 M SnCl_2 , and 0.1 M $\text{SnCl}_2 + 0.1 \text{ M SbCl}_3$ in $[\text{Py}_{1,4}]\text{TfO}$ recorded on copper substrates (without using PC templates). Scan rate 10 mV/s



various copper–tin alloys can occur such as Cu_6Sn , Cu_3Sn , and Cu_4Sn [36]. The CV of 0.1 M $\text{SbCl}_3 + 0.1 \text{ M SnCl}_2$ in $[\text{Py}_{1,4}]\text{TfO}$ displays three processes in the forward scan with the consequent anodic ones in the reverse scan. The cathodic processes c_1 and c_2 can be ascribed to surface alloying and bulk deposition of Sn, respectively, whereas c_3 can be correlated to the deposition of SbSn. The peaks a_1 and a_2 recorded on the anodic scan can be attributed to the stripping of Sn and SbSn, respectively.

It should be noted that the shape of the recorded cathodic and anodic waves is not typical for conventional reduction or oxidation peaks which are normally obtained in aqueous electrolytes. This is not surprising as in ionic liquid electrolytes conventional double layers do not form at the electrode/electrolyte interface. Instead of the formation of a simple double layer, multilayers are formed and the number of layers is dependent on the ionic liquid species as well as the surface properties of the electrode [37]. The formation of such interfacial solvation layers can simply influence the electrochemical behavior and the shape of the observed processes.

Constant potential electrolysis was performed to obtain Sb and Sn–Sb deposits from the employed electrolyte on

the copper substrate surface for 1 h. After the potentiostatic electrodeposition experiments, the deposit was washed with isopropanol to remove ionic liquid residues and analyzed by SEM–EDX.

Figure 2a shows the SEM micrograph of an electrodeposited antimony film on copper substrate obtained at -1.5 V for 1 h (total charge 432 C) from 0.1 M $\text{SbCl}_3/[\text{Py}_{1,4}]\text{TfO}$. As seen from the SEM micrograph is a dense deposit with some dendrites on the top of the homogeneous Sb film underneath. This indicates the formation of a smooth Sb film at the beginning of the deposition experiment and by the ongoing time Sb dendrites formed. The EDX profile of the deposited Sb film shows the existence of antimony besides C, N, S, and F from the ionic liquid as depicted in Fig. 2b. The elemental composition of the obtained film is shown in the inset of Fig. 2b.

To synthesize Sb–Sn alloy films on copper from 0.1 M $\text{SnCl}_2 + 0.1 \text{ M SbCl}_3/[\text{Py}_{1,4}]\text{TfO}$, potentiostatic electrodeposition was carried out at a potential of -1.85 V (vs. Pt) (total charge 648 C). The SEM image of the obtained deposit is presented in Fig. 3a. As seen, the deposit is highly dense, and it contains spherical particles with sizes in the micrometer range. The presence of solid impurities in Fig. 3 could

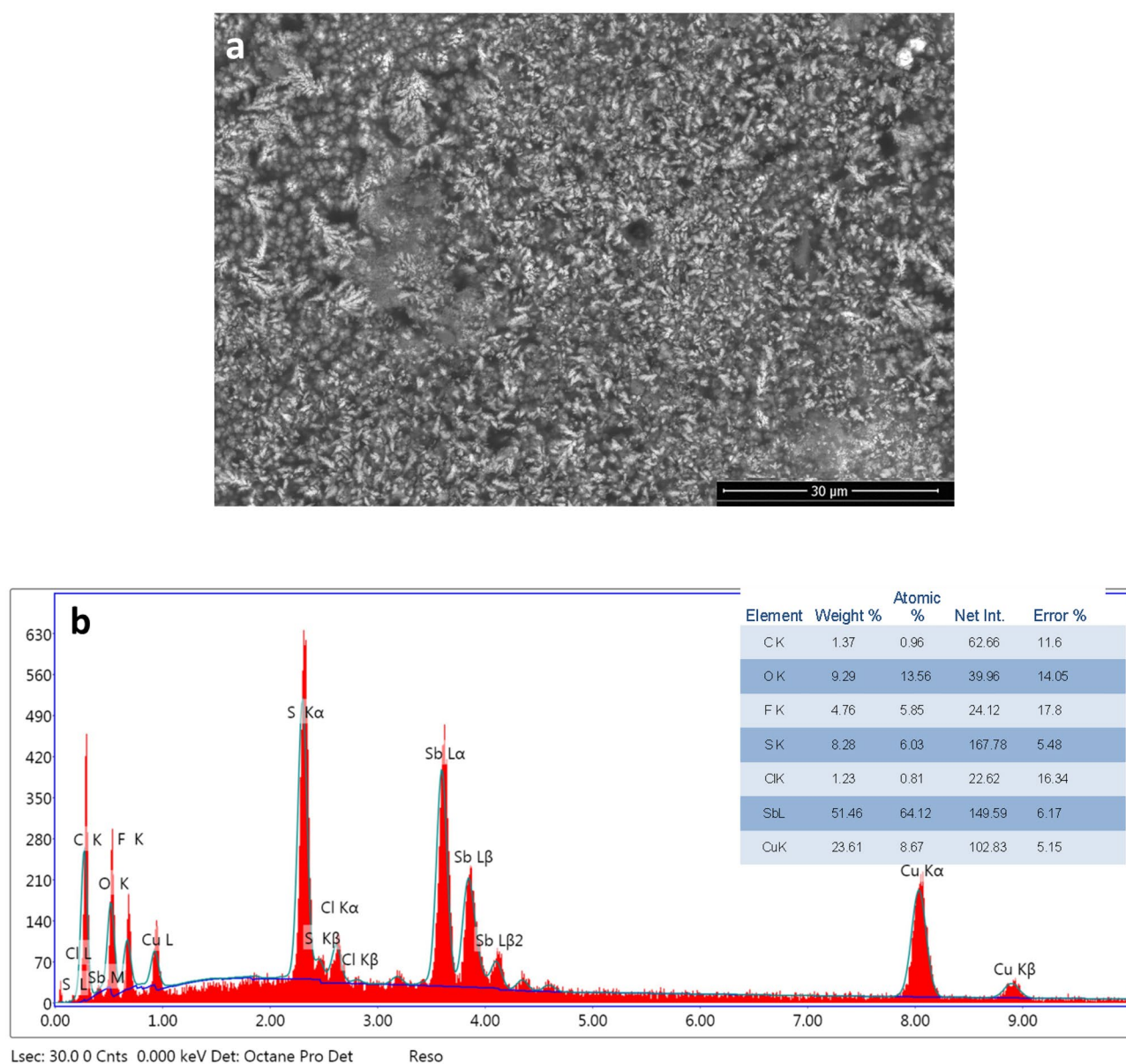


Fig. 2 **a** SEM of the electrodeposited Sb film obtained by potentiostatic deposition from 0.1 M SbCl_3 in $[\text{Py}_{1,4}]\text{TfO}$ at -1.5 V. **b** EDX profile of the electrodeposited Sb film. Inset: the elemental composition

be attributable to the trapped electrolyte residues which might be hydrolyzed during rinsing. The EDX spectrum of the electrodeposited layer is shown in Fig. 3b. The EDX spectrum shows the distinctive peaks of Sn and Sb signifying the co-deposition of Sb and Sn at the applied potential. The weight ratio of Sn/Sb in the obtained deposit was estimated to be 0.7/0.3. Peaks for the ionic liquid residues are also observed revealing the presence of trapped ionic liquid species in the obtained Sn-Sb deposit.

The XRD patterns of the electrodeposited Sb, Sn, and Sb-Sn films obtained on copper are shown in Fig. 4. The XRD patterns of the electrodeposited Sb films show

pronounced diffraction peaks at $2\theta = 28.8^\circ$, 42.16° , 59.6° , and 68.9° . The recorded diffraction peaks are well indexed to (0 1 2), (1 1 0), (0 2 4), and (1 2 2) diffraction peaks of the face-centered cubic (fcc) crystalline structure of antimony (JCPDS File No. 35-0732). The broadening of the peak recorded at 28.8° reveals the formation of very fine Sb crystallites [38]. The appearance of the characteristic diffraction peaks of the Cu substrate signifies that the electrodeposited Sb film is not thick enough to conceal the substrate. The diffraction patterns of the electrodeposited Sn film display weak diffraction peaks at 2θ values of 31.4° , 32.7° , 47.4° , and 54.4° , which are assigned to the (200), (101), (211), and

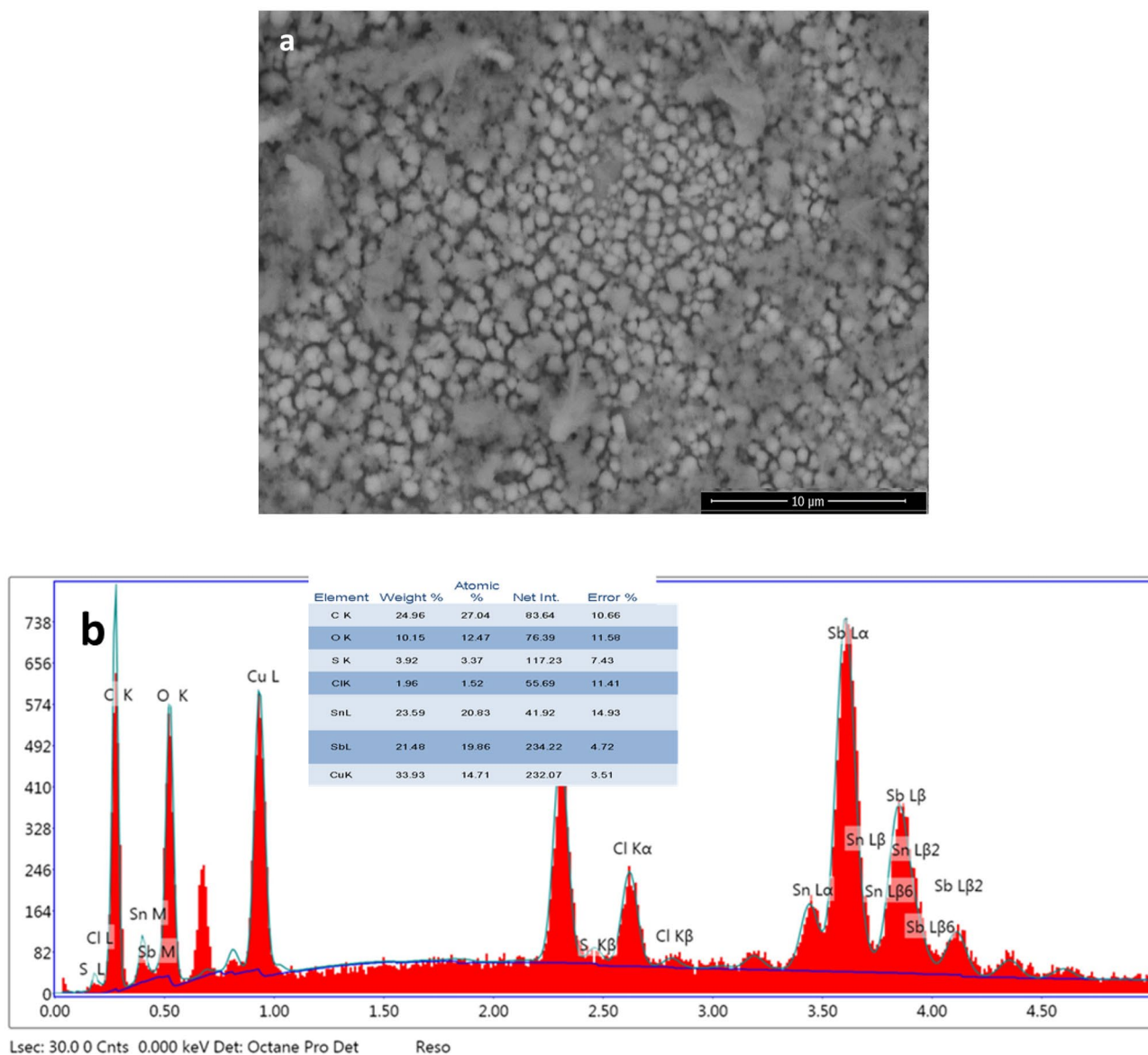


Fig. 3 **a** SEM image of Sb-Sn deposit obtained by potentiostatic deposition from 0.1 M SbCl_3 + 0.1 M SnCl_2 in $[\text{Py}_{1,4}]\text{TfO}$ at -1.85 V (vs. Pt). **b** EDX spectrum of the obtained deposit. Inset: the elemental composition

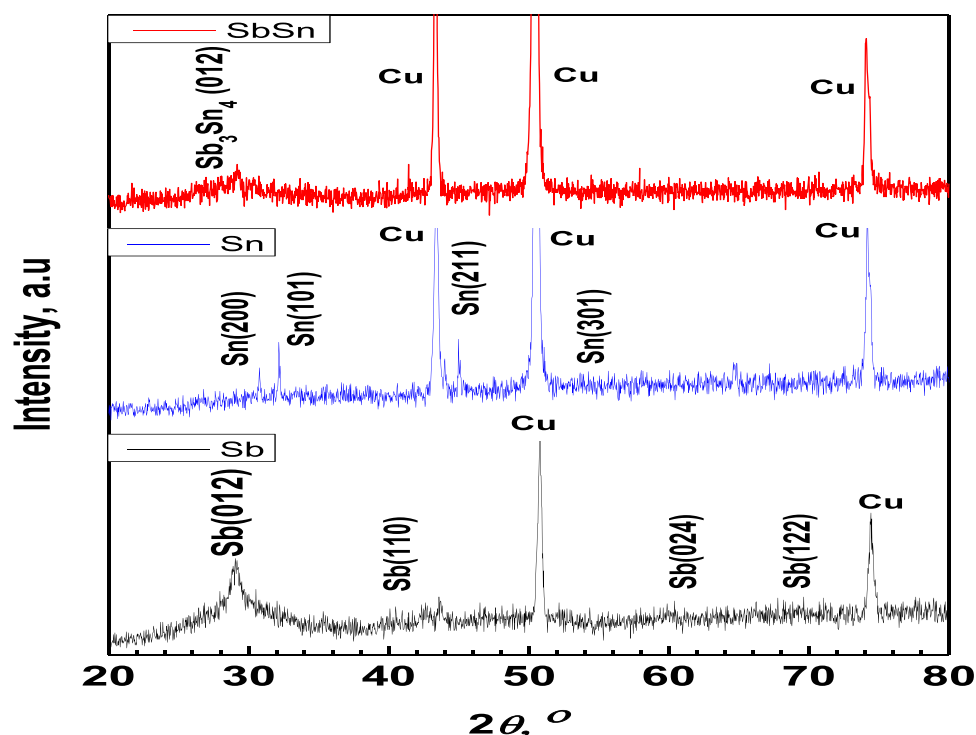
(301) planes of the tetragonal tin, respectively (JCPDS File No. 04-0673) [36]. The XRD patterns of the electrodeposited Sb-Sn film show, besides the Cu diffraction peaks, weak diffraction peaks of Sb_3Sn_4 alloy (JCPDS File No. 33-0118). This indicates the electrodeposition of a very thin Sb_3Sn_4 alloy film on the Cu substrate.

Nanowire arrays of Sb, Sn, and Sb-Sn were synthesized in $[\text{Py}_{1,4}]\text{TfO}$ by a template-assisted electrodeposition technique. First, the nanowires grow within the pores of the polycarbonate PC membrane. Subsequently, a Cu film was electrodeposited on the back side of the PC membrane (the Au-sputtered side) as a supporting layer to make the nanowire freestanding and mechanically stable.

The electrochemical deposition of the copper-supported layer is essential to prevent the collapsing of the nanowire structure after the chemical dissolution of the PC membrane. The collapse of the nanowire structure is a notable drawback when the nanowire electrode is employed as an anode for Li-ion batteries. Most of the anode material would be electrochemically inactive as a result of poor contact with the current collector. Hence, the formation of a thick copper supporting layer is beneficial.

We tried first to electrodeposit a thick layer of Sb on the Au-sputtered side of the membrane to serve as a backing layer for the electrodeposited nanowires. The SEM image of Fig. 5a shows a top view of the Au-sputtered side of the track-etched

Fig. 4 XRD patterns of the electrodeposited Sb, Sn, and Sb-Sn alloy films



polycarbonate membrane prior to the electrodeposition. The pores are distributed randomly in the membrane and doublet or triplet pores exist too. The membrane has a pore diameter of 400 nm. A relatively thick layer of Sb was electrodeposited on the back side of the PC membrane from the ionic liquid electrolyte 0.1 M $\text{SbCl}_3/[\text{Py}_{1,4}]\text{TfO}$ at -1.6 V (vs. Pt) for 1 h. The SEM image of Fig. 5b shows the morphology of the obtained Sb layer. As seen, a dendritic structure is formed which makes this layer unsuitable as a backing layer as it is brittle. Thus, a thick Cu layer was electrodeposited on the sputtered side of the gold sputtered membrane instead of Sb to act as a supporting layer for the nanowire structure. The electrodeposited copper layer is compact and dense as shown in Fig. 5c. Hence, this layer can dually function as a supporting layer and as a current collector when the nanowire electrode will be employed in Li-ion batteries. Therefore, to obtain freestanding nanowires, a two-step electrodeposition regime was developed. In the first step, the nanowires were grown in the pores of the non-sputtered side of the membranes. Then, a thick copper film with thickness of about $5\ \mu\text{m}$ was electrodeposited on the back side of the membrane.

Figure 6 shows the SEM images of freestanding Sb nanowire arrays fabricated by potentiostatic electrodeposition at a potential of -1.6 V (vs. Pt) for 1 h in 0.1 M $\text{SbCl}_3/[\text{Py}_{1,4}]\text{TfO}$. The SEM images were taken after removing the template by chemical dissolution in dichloromethane. The SEM image of Fig. 6 shows the homogeneity of the growing Sb nanowires over the surface which reveals the excellent wettability of the employed ionic liquid electrolyte as

obviously all pores were filled with the electrolyte. This can be regarded as another advantage of the employing ionic liquid electrolytes instead of aqueous ones as the surface tension of water is higher than those of ionic liquids which, in turn, leads to a bad wettability of aqueous electrolytes. The higher magnification SEM image (inset of Fig. 6) clearly reveals the high quality of the obtained Sb nanowire films with a perfect adhesion to the supporting layer. The diameter of the obtained Sb nanowires was found to be about 400 nm which is consistent with the nominal pore diameter of the employed PC membrane.

A nanowire film of Sn obtained on the electrodeposited Cu supporting layer in 0.1 M SnCl_2 in $[\text{Py}_{1,4}]\text{TfO}$ at -1.25 V (vs. Pt) for 1 h is shown in Fig. 7. The nanowires are homogeneously formed all over the surface and they show a good adhesion to the Cu-supporting layer, as seen in the higher magnification SEM image of Fig. 7b.

Freestanding Sb-Sn alloy nanowires were also obtained from a solution of 0.1 M $\text{SnCl}_2 + 0.1$ M SbCl_3 in $[\text{Py}_{1,4}]\text{TfO}$ by electrodeposition into gold sputtered polycarbonate membranes having pore sizes of 400 nm. Figure 8 demonstrates a representative SEM micrograph of the freestanding Sn-Sb alloy nanowires electrodeposited potentiostatically at -1.85 V for 1 h. It is clearly seen that the obtained nanowires are leveled with an even distribution of the nanowires all over the surface. The higher magnification SEM image (inset of Fig. 8) nicely shows the high quality of the obtained freestanding Sn-Sb nanowire arrays with an excellent contact with the supporting Cu layer.

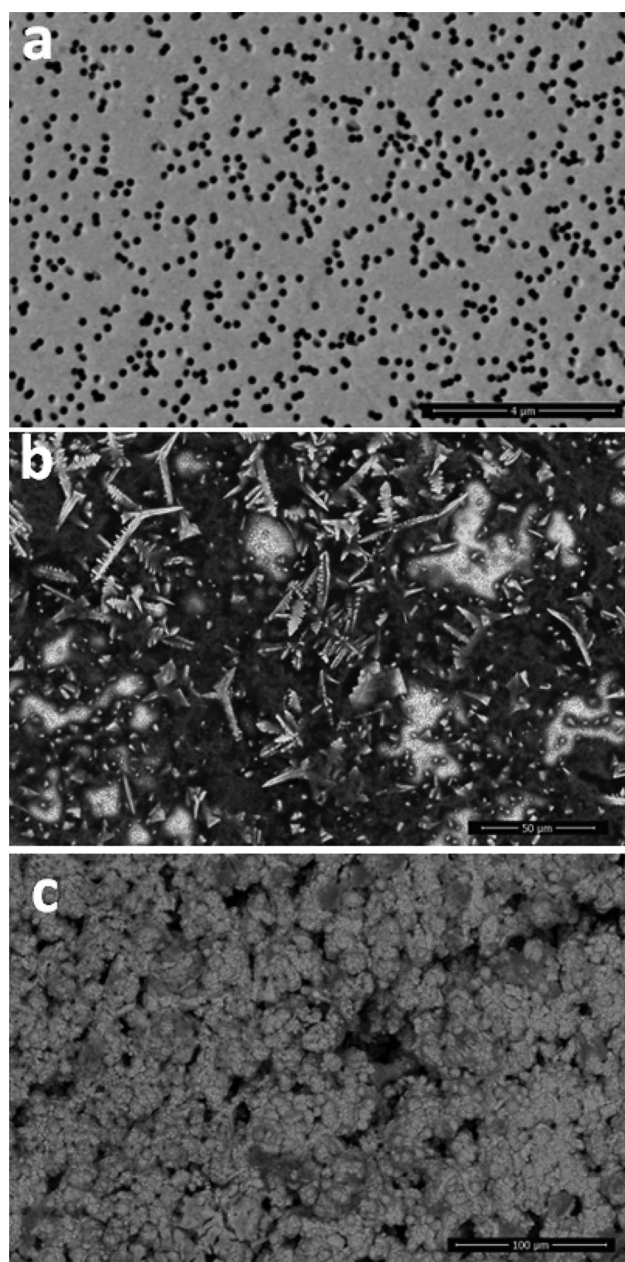


Fig. 5 SEM images of **a** gold sputtered PC membrane. **b** Sb thin film deposited from 0.1 M SbCl_3 in $[\text{Py}_{1,4}]\text{TfO}$ on the gold sputtered membrane at -1.5 V for 1 h. **c** Cu film electrodeposited on the gold sputtered side of the membrane

Electrochemical performance of electrodeposited nanowire electrodes

In order to explore the Li storage capacity of the electrodeposited Sn, Sb, and SbSn nanowire films, preliminary galvanostatic charge/discharge characteristics and initial cycle performance were investigated. Figure 9a depicts the charge/discharge profiles for the 50th galvanostatic cycles at a current density of $100 \mu\text{A}$ and a potential

range from 5 mV to 1.5 V (vs. Li/Li^+). The charge capacity of the electrodeposited Sb nanowires in the first cycle was found to be 580 mA h/g , which is comparable to the theoretical capacity of Sb (660 mA h/g) [10], and also about a factor 1.5 higher than that of the conventional graphite anode (372 mA h/g) [11]. However, the second discharge specific capacities and charge specific capacities are about 480 and 447 mA h/g , respectively. It is observed from the voltage profile of Sb nanowire arrays deposited from $[\text{Py}_{1,4}]\text{TfO}$ ionic liquid that there are some potential plateaus observed during the discharge-charge processes, which can be attributed to the significant lithiation-delithiation reactions on the Sb NWs. After the 50th cycles, the discharge and charge specific capacities of the Sb electrodeposit were found to be 338 and 340 mA h/g , respectively as shown in Fig. 9a. High irreversible capacity loss between the first and second discharge is generally attributed to the SEI layer formation and the irreversible insertion of Li^+ ions into free spaces present on the Sb NWs surface.

Figure 9b shows the voltage profiles of the Sn nanowire films in different cycles at a current density of $100 \mu\text{A/g}$ between 0.01 and 2.0 V potential window (vs. Li/Li^+). The obtained specific capacity of the electrodeposited Sn nanowire electrode was found to be 744 mA h/g in the first cycle which is comparable to the theoretical specific capacity of Sn anode (992 mAh/g) [9]. As shown in Fig. 9b, there are some potential plateaus observed during the discharge-charge processes, which can be attributed to the significant lithiation-delithiation reactions on the Sn electrodeposit. Reaching the 50th cycles, the discharge and charge specific capacities of the Sn electrodeposit were found to be 283 and 285 mA h/g , respectively. The observed high irreversible capacity loss on cycling might be attributable to the disintegration of the nanowire arrays. Figure 9c displays the cycling profile of the electrodeposited SbSn nanowire electrode in the voltage range between 0.01 and 2.0 V for fifty cycles. The charge-discharge performance of SbSn nanowires electrode exhibits several plateaus, which are characteristic of the electrochemical alloying of SbSn and Li. The capacity was found to be about 800 mA h/g in the first cycle, and after the 50th cycle, the capacity dropped to about 493 mA h/g . The SbSn nanowire film did not exhibit a high capacity loss on cycling as in the case of Sn nanowire films, signifying the good cyclic performance of the SbSn nanowire electrode. The first results of the evaluation of the cyclic performance of the synthesized nanowire electrodes indicate the possibility of the application of the synthesized electrodes as anodes for future generation of the Li-ion batteries. Further work is needed for comprehensive characterization of the electrochemical performance of the fabricated nanowire electrodes.

Fig. 6 Top-view SEM images of Sb nanowires at different magnifications obtained potentiostatically from 0.1 M SbCl_3 in $[\text{Py}_{1,4}]\text{TfO}$ at potential of -1.6 V vs. Pt for 1 h. Inset: high-magnification SEM image of the nanowires with diameter of approximately 400 nm

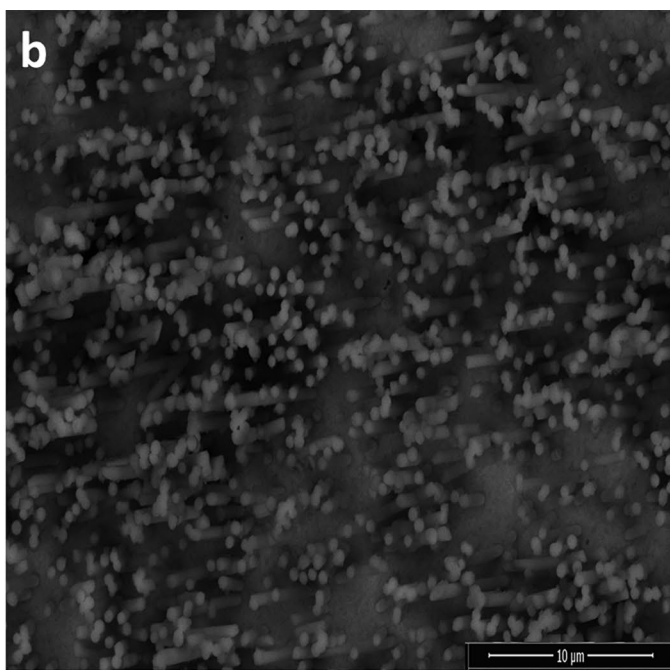
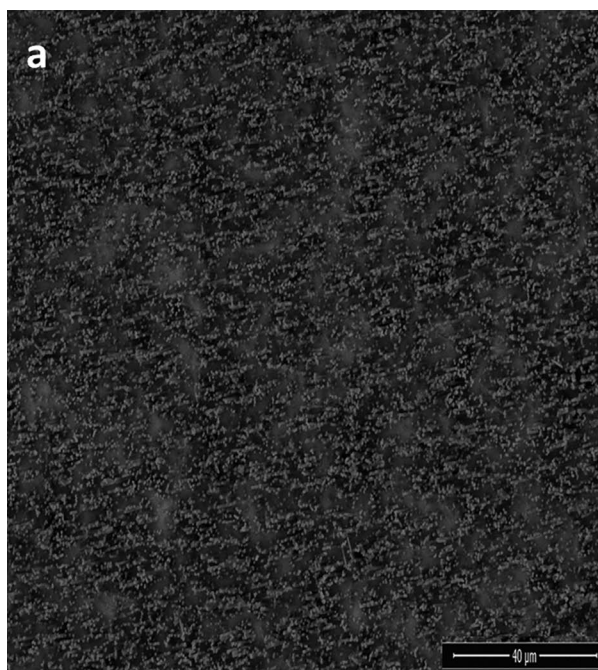
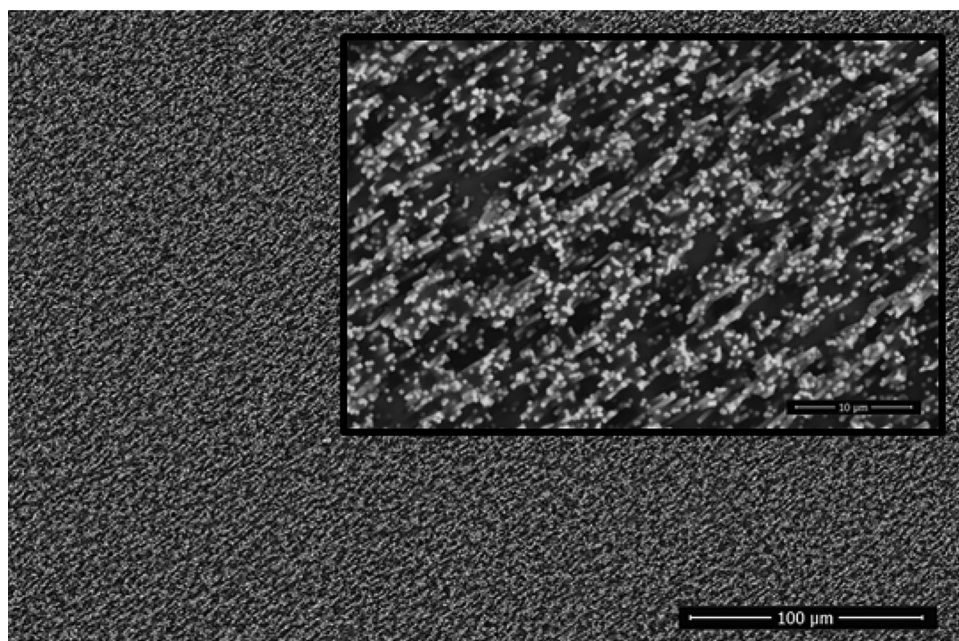


Fig. 7 **a** Top-view SEM images of Sn nanowires at different magnifications obtained potentiostatically from 0.1 M SnCl_2 in $[\text{Py}_{1,4}]\text{TfO}$ at a potential of -1.25 V vs. Pt for 1 h. **b** High-magnification SEM image of the nanowires with a diameter of approximately 400 nm

Fig. 8 Top-view SEM images of Sb-Sn nanowires obtained potentiostatically from 0.1 M $\text{SnCl}_2 + 0.1 \text{ M SbCl}_3$ in $[\text{Py}_{1,4}]\text{TfO}$ at potential of -1.85 V vs. Pt for 1 h. Inset: high-magnification SEM image shows the freestanding nanowires

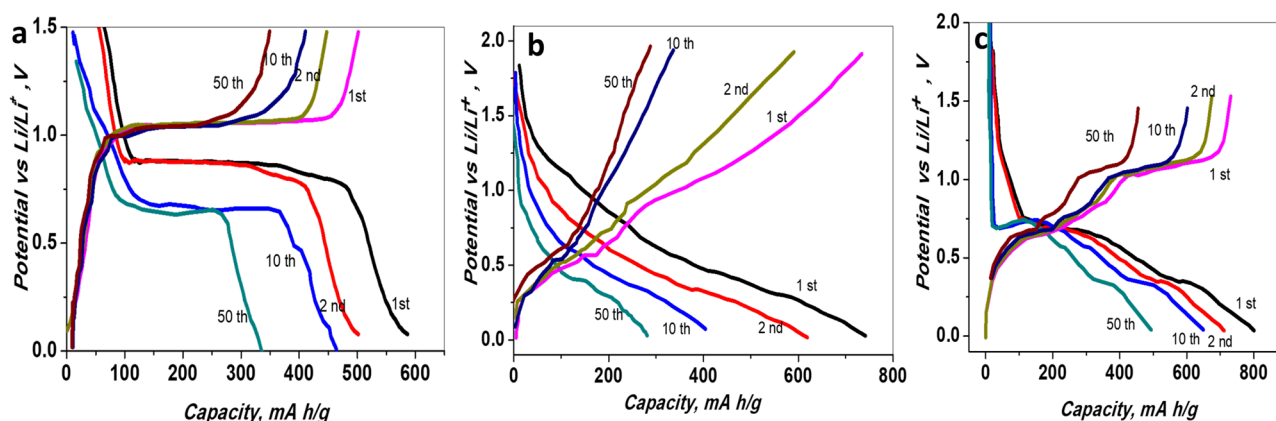
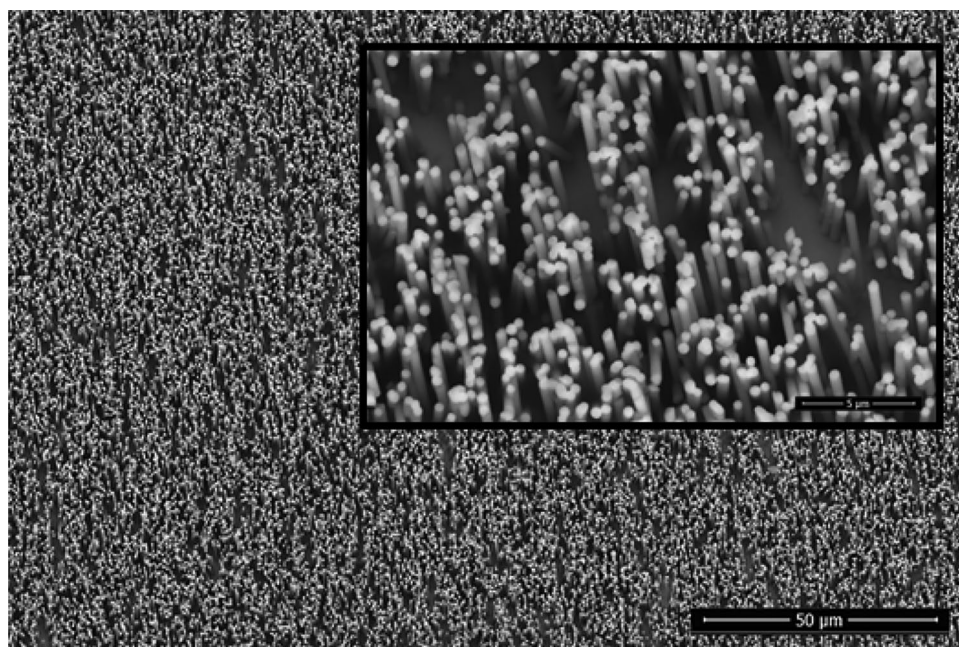


Fig. 9 Galvanostatic charge/discharge profiles of the electrodeposited nanowire films of **a** Sb, **b** Sn, and **c** SbSn obtained from $[\text{Py}_{1,4}]\text{TfO}$

Conclusion

We have presented the electrochemical deposition of free-standing nanowire arrays of Sn, Sb, and of SbSn via a template-assisted approach in the ionic liquid $[\text{Py}_{1,4}]\text{TfO}$. A two-step electrodeposition regime was developed for the fabrication of strongly adherent, freestanding nanowire arrays on an electrodeposited copper supporting layer. In the first step, the nanowires were grown in the pores of the template. Then, a thick copper film was electrodeposited on the back side of the membrane. This layer can dually function as a supporting layer and as a current collector when the fabricated nanowire electrodes will be employed in Li-ion

batteries. The capacitance and cycle stability of the synthesized nanowire films were investigated. It was found that the obtained nanowire films are promising anode candidates for the future generation of Li-ion batteries.

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Data availability Correspondence and requests for materials should be addressed to S. Zein El Abedin. All data generated or analyzed during this study are included in this article.

Declarations

Competing interests The authors declare no competing interests.

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